A Comparative Study of Evaluation Methodologies for Anomaly Detection on ICS Benchmarks*

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Abstract

In Industrial Control System (ICS) anomaly detection research, benchmark datasets collected from testbeds such as SWaT, WADI, and HAI have been widely used; however, concerns about the reliability of reported performance have been raised persistently due to inherent limitations in both the datasets and the evaluation methodologies. In particular, widely used evaluation metrics such as Point-Adjusted F1 have been shown to overestimate model performance, and fixed thresholds fail to capture data characteristics. However, it has not been systematically examined whether the choice of evaluation methodology influences model comparison outcomes on ICS benchmarks. To address this gap, we compared the performance of traditional ML models (PCA, Isolation Forest) and state-of-the-art DL models (USAD, TranAD) across four ICS benchmark datasets: SWaT, WADI, HAI 21.03, and HAI 23.05. In Exp1, we applied a variety of evaluation metrics (AUROC, AUPRC, Affiliation Precision/Recall) to identical anomaly scores. In Exp2, we applied Oracle and Label-free threshold methods (Best F1, NDT, P_{95}, P_{99}) to quantify how the choice of evaluation methodology shifts relative model rankings. The results show that ML models matched or outperformed DL models in 15 out of 16 dataset-metric combinations. Moreover, the PA-F1 score of a single model varied by up to 0.719 across threshold methods, confirming that the choice of evaluation methodology can considerably alter model rankings. These results indicate that ICS anomaly detection research requires not only improvements in detection models but also systematic examination of evaluation methodologies.

1 Introduction

Industrial Control Systems (ICS) monitor and automatically regulate the physical processes of industrial infrastructure such as power plants, water treatment facilities, and manufacturing processes through sensors and control devices. As a form of Cyber-Physical System (CPS) in which the cyber and physical domains are tightly coupled, any incident within an ICS can directly translate into injuries or damage to facilities, which has motivated a growing body of research on cyber-resilience and safe operation [18]. In this context, time-series anomaly detection for the early identification of cyber attacks, such as sensor value manipulation or unauthorized actuator control, has emerged as a central topic; datasets collected from testbeds that emulate industrial processes and incorporate injected attack scenarios have been widely

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used as primary benchmarks for this purpose. Autoencoder- and Transformer-based deep learning models are widely adopted as detection methods and have achieved high performance on these benchmarks [4].

However, recent studies reveal that time-series anomaly detection benchmarks suffer from intrinsic limitations such as labeling errors and unrealistic anomaly ratios [27], and similar issues have been documented in ICS benchmarks [4]. Reproducing existing models under identical conditions has been reported to yield slightly lower performance than originally claimed [9]. In addition, recent models have been observed to degrade under noise-perturbed settings [15]. These findings raise concerns about whether reported results can be reliably reproduced. Regarding evaluation methodology, recent studies criticize the Point-Adjusted F1 (PA-F1) for inflating model performance [5, 10], and prior work identifies fixed thresholds as inadequate for capturing data characteristics [11]. A variety of alternative metrics and threshold schemes have been proposed to address these limitations, yet whether these alternatives alter model comparisons on the same ICS benchmarks has not been systematically examined.

Motivated by this gap, we examine the reliability of anomaly detection evaluation on ICS benchmark datasets. The main contributions are summarized as follows:

- We compare traditional machine learning (ML) models and recent deep learning (DL) models under a variety of evaluation metrics on four ICS benchmark datasets (Exp1).
- Under identical anomaly scores, we compare performance across different threshold schemes and examine how the choice of evaluation methodology shifts model rankings (Exp2).
- We extend the analysis to HAI 21.03 and HAI 23.05, two ICS benchmarks that have received comparatively little scrutiny, and further analyze how differences in data composition between two versions of the same testbed influence detection results.

2 Related Work

2.1 Anomaly Detection Methods

Time-series anomaly detection has long relied on traditional machine learning methods. Principal Component Analysis (PCA)-based anomaly detection, which measures reconstruction error based on learned correlations among multiple sensor values [3], has been applied to industrial process monitoring, and Isolation Forest (IForest) [12], which isolates anomalies by randomly partitioning data, is a foundational method in tree-based anomaly detection. These traditional methods have been widely adopted in industrial settings, as they provide stable performance without requiring DL training.

Subsequently, researchers have introduced a variety of DL approaches. Hundman et al. [7] proposed an LSTM-based model as a prediction-based approach, which predicts normal sequences and uses prediction error as the anomaly score. In the reconstruction-based category, USAD [2], which combines an autoencoder with adversarial training, is a notable example. As a generative approach, Su et al. [22] combined a stochastic RNN with a VAE in OmniAnomaly. Among Transformer-based approaches, TranAD [24], which captures the full sequence context and applies adversarial training, is representative, and subsequent work introduced models that leverage inter-variable attention mechanisms [8]. Graph-based approaches that explicitly model spatial dependencies among sensors [29] have also emerged for environments such as ICS where multiple sensors interact. Among these, USAD and TranAD achieved state-of-the-art performance on major benchmarks including SWaT and WADI, and became prominent DL anomaly

detection models. However, Sarfraz et al. [17] showed that traditional ML methods, for example, PCA reconstruction error alone, can outperform existing DL models on SWaT and WADI. This suggests that the performance advantage of DL models may vary across datasets and evaluation settings.

These findings, however, rest primarily on SWaT and WADI, and whether the same trend holds on other ICS datasets remains underexplored. In this study, we selected PCA and IForest as traditional models and USAD and TranAD as DL methods, and compared the performance of the two model types on four ICS datasets, including not only SWaT and WADI but also HAI 21.03 and HAI 23.05.

2.2 Benchmark Datasets

In time-series anomaly detection research on ICS, prior work has primarily used datasets collected from testbeds with injected attack scenarios [4]. Among the most prominent are SWaT [13] from a water treatment testbed and WADI [1] from a water distribution testbed, released by the iTrust laboratory, and HAI [19] from a power generation testbed, collected by ETRI, which integrates boiler, turbine, and water treatment processes through a Hardware-in-the-Loop (HIL) simulator.

However, several studies have identified various limitations in time-series anomaly detection benchmarks in general. Wu and Keogh [27] demonstrated that benchmarks released by major organizations such as Yahoo, Numenta, and NASA suffer from four types of flaws: trivial problems solvable by simple methods (triviality), unrealistic anomaly ratios, mislabeled ground truth, and run-to-failure bias, in which anomalies are concentrated in the later portion of the data. For ICS datasets specifically, Correia et al. [4] found that SWaT contains numerous anomaly cases detectable by the single feature alone, and WADI includes many features composed of missing values or constant values, posing challenges for evaluation. Wenig et al. [26] further noted that existing multivariate benchmarks contain anomalies that are effectively detectable at the univariate level, failing to demonstrate the strengths of multivariate algorithms.

In contrast, the HAI dataset, while also incorporating attack scenarios on an HIL-based testbed, has not been sufficiently examined in this regard. In this study, we included HAI 21.03 and HAI 23.05 in addition to SWaT and WADI, to examine whether similar patterns hold on ICS datasets for which such analyses remain scarce.

2.3 Performance Evaluation

In time-series anomaly detection, F1-score has served as the primary performance metric; to enable segment-level evaluation, Xu et al. [28] proposed Point-Adjusted F1 (PA-F1), which treats an entire anomaly segment as correctly detected if even a single time point within it is identified, and it has since become widely adopted. However, multiple studies have raised persistent concerns that PA-F1 inflates model performance. Kim et al. [10] mathematically demonstrated that PA-F1 scores converge to 1 in proportion to the length of the anomaly segment, and Sehili and Zhang [5] showed that selecting only a small number of random time points from SWaT and WADI yields PA-F1 scores approaching 0.95. Kang and Kang [8] documented a performance gap of up to 0.48 when PA-F1 and other metrics are applied to the same model. In addition, while most studies measure performance based on a fixed threshold, prior work has noted that fixed thresholds fail to capture the distributional characteristics of the data, as the choice of threshold can substantially alter evaluation outcomes [11].

To address these limitations, researchers have proposed various alternatives to fixed threshold selection and point-wise evaluation that assesses individual time points independently.

Range-based precision/recall [23] evaluates performance at the segment level rather than at individual time points; Volume Under the Surface (VUS-ROC, VUS-PR) [14] integrates the area under ROC/PR curves across multiple window sizes to provide evaluation that is robust to temporal offset; and Affiliation Precision/Recall (Aff_P, Aff_R) [6] converts the temporal distance between predicted time points and actual anomaly segments into probability-based scores, enabling parameter-free evaluation. For threshold selection, Hundman et al. [7] introduced NDT, which dynamically sets thresholds based on data characteristics.

However, whether these improvements in evaluation metrics and threshold methods are effective on ICS benchmark datasets where issues have been documented has not been sufficiently verified. In this study, holding the model output fixed, threshold-free metrics (AUROC, AUPRC) and distance-based metrics (Aff_P, Aff_R) were applied in Exp1 to compare performance across models from multiple perspectives, while five threshold schemes, including Oracle methods (Best F1, Best $F1_{PA}$) and Label-free methods (NDT, P_{95}, P_{99}), were applied in Exp2 to analyze how the choice of evaluation methodology influences relative model performance.

3 Methodology

3.1 Models

Principal Component Analysis (PCA). PCA is a technique that extracts principal components from high-dimensional data to reduce its dimensionality, and is used for data visualization, noise removal, and feature extraction. In anomaly detection, test data is projected onto the principal component subspace learned from normal data and then reconstructed, and the reconstruction error, defined as the difference between the reconstructed and original data, is used as the anomaly score. Data that deviates from normal patterns yields higher reconstruction error due to inaccurate reconstruction. This approach has long been used in Multivariate Statistical Process Control (MSPC) for monitoring multiple sensor values and detecting process anomalies [3], and has recently been applied to time-series anomaly detection, achieving performance equal to or higher than that of recent DL models on SWaT and WADI [17]. We used the experimental resources from that work [17], deriving the reconstruction error for each feature j as $e_j = |x_j - \hat{x}_j|$ and taking the maximum score across all features as the final anomaly score $s = \max_j e_j$. Since a single feature with the largest error determines the score, PCA operates at the individual feature level during scoring, despite capturing inter-variable correlations during training.

Isolation Forest (IForest). IForest is a tree-based unsupervised algorithm that detects anomalies by randomly partitioning data points [12]. Each isolation tree repeatedly selects a random feature and split value to partition the data; since anomalous points can be isolated with fewer partitions than normal data, they exhibit shorter path lengths. The final anomaly score is derived by averaging and normalizing these path lengths across multiple trees, where a shorter path length indicates that the data point was easily isolated and is thus more likely to be anomalous. Like PCA, IForest operates on individual feature splits rather than explicitly modeling inter-variable correlations. This is a traditional ML method that has been widely used in anomaly detection, and we used the scikit-learn library [16].

UnSupervised Anomaly Detection on Multivariate Time Series (USAD). USAD is an unsupervised anomaly detection model that combines an autoencoder with adversarial training inspired by GANs [2], consisting of a shared encoder E and two decoders D_1 and

D_2 that form two autoencoders $AE_1(E + D_1)$ and $AE_2(E + D_2)$. Training proceeds in two phases: in Phase 1, each autoencoder minimizes the reconstruction error of the input window W to learn normal patterns, and in Phase 2, adversarial training between the two autoencoders further improves the reconstruction accuracy for normal data. The anomaly score at test time is derived as a weighted combination of the reconstruction errors from both autoencoders, as shown in Equation (1), where α and β are the weights satisfying $\alpha + \beta = 1$. In this study, we set $\alpha = \beta = 0.5$.

$$s = \alpha\|W - AE_1(W)\|_2 + \beta\|W - AE_2(AE_1(W))\|_2 \quad (1)$$

TranAD. TranAD is a Transformer-based anomaly detection model [24] that reconstructs the input window using a Transformer encoder and two decoders. As in USAD, training proceeds in two phases, where self-conditioning that uses the reconstruction error from Phase 1 as a focus score concentrates detection sensitivity on sub-sequences that are poorly reconstructed, while adversarial training between the two decoders simultaneously improves reconstruction accuracy for normal data. The anomaly score at test time is derived by averaging the reconstruction errors of the two decoder outputs, with larger values indicating greater deviation from normal patterns. We used the resources released in that work [25].

3.2 Datasets

Secure Water Treatment (SWaT). SWaT is an ICS security dataset collected from a water treatment testbed built by the iTrust laboratory at the Singapore University of Technology and Design [13], and is one of the most widely used benchmarks in ICS anomaly detection research. Data from 51 sensors and actuators were collected from a scaled-down physical testbed that emulates an industrial environment, consisting of 7 days of normal operation and 4 days of attack scenarios. The attack scenarios include 36 cases such as sensor value manipulation and actuator control, and the anomaly ratio of the test data is approximately 12.1%.

Water Distribution (WADI). WADI is a dataset from a water distribution testbed built by the same institution as SWaT [1], with data collected from 127 sensors and actuators. We excluded 4 features whose values are entirely missing and used the remaining 123 features for the experiments. The dataset consists of 14 days of normal operation data and 2 days of attack scenario data, including 15 attack scenarios. The anomaly ratio of the test data is approximately 5.8%, lower than that of SWaT.

HIL-based Augmented ICS (HAI). HIL-based Augmented ICS is an ICS security dataset built by the Electronics and Telecommunications Research Institute (ETRI), with the testbed consisting of three physical processes: boiler (P1), turbine (P2), and water treatment (P3), connected through a Hardware-in-the-Loop (HIL) simulator (P4) [19]. Each of the three processes is controlled by a different industrial controller, and the HIL simulator emulates steam turbine generation and pumped-storage hydropower generation so that the three processes operate as an integrated power generation system. Compared to established benchmarks such as SWaT and WADI, this dataset has been less frequently used in anomaly detection research; accordingly, we included two versions, HAI 21.03 [20] and HAI 23.05 [21], to assess performance across a broader range of ICS environments.

HAI 21.03 contains 79 features and 50 attack scenarios, with an anomaly ratio of approximately 2.2% in the test data. From HAI 22.04 onward, attack scenarios became more sophisticated, with detection difficulty reported to be approximately four times that of HAI 21.03 [21].

HAI 23.05, the latest version in this series, consists of 86 features and 52 attack scenarios, with an anomaly ratio of approximately 4.00% in the test data. By comparing the 21.03 and 23.05 versions from the same testbed, we analyzed how changes in dataset composition influence detection results.

3.3 Evaluation Metrics

In this section, evaluation metrics are categorized into three groups based on their characteristics.

Point-Adjusted F1 (PA-F1). F1-score is the harmonic mean of Precision and Recall, and is the most widely used performance metric in classification tasks including anomaly detection. The formulas for deriving this score are as follows:

$$Precision(P) = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad Recall(R) = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \quad F_1 = \frac{2PR}{P+R} \tag{2}$$

where TP (True Positive) is the number of time points where anomalies are correctly identified as anomalies, FP (False Positive) where normal points are incorrectly identified as anomalies, and FN (False Negative) where anomalies are incorrectly identified as normal. In time-series anomaly detection, Point-Adjusted F1 (PA-F1) was proposed as an extension for segment-level evaluation [28]. PA-F1 applies point adjustment, treating the entire anomaly segment as successfully detected if at least one time point within it is correctly identified, and was designed from the perspective that detecting the occurrence of an anomalous event itself is important. However, this metric has been criticized for overestimating actual detection performance [5,10].

Threshold-free Metrics. Since F1-score and PA-F1 require setting a specific threshold to binarize anomaly scores into normal and anomalous classifications before evaluation, performance can vary substantially depending on the threshold selection. Threshold-free metrics address this dependency by evaluating performance comprehensively across all possible thresholds using the anomaly scores directly, and are also referred to as threshold-independent measures [14]. AUROC (Area Under Receiver Operating Characteristic) evaluates performance by varying the threshold, computing the True Positive Rate and False Positive Rate at each point, plotting these as a curve, and measuring the area under it. AUPRC (Area Under Precision-Recall Curve) similarly computes the area under the curve of the Precision-Recall relationship. Since the computation of AUROC involves True Negatives in the False Positive Rate (FPR = FP/(FP+TN)), performance can be overestimated on datasets with low anomaly ratios, as the overwhelming number of normal data points keeps the FPR low. Since the Precision in AUPRC does not involve True Negatives, it can mitigate this limitation.

Distance-based Metrics. Aff_P and Aff_R address this limitation using the distance between a predicted point x and a ground truth segment:

$$d(x, J) = \begin{cases} 0 & \text{if } x \in J \\ \min(|x - j_{\mathrm{start}}|,\ |x - j_{\mathrm{end}}|) & \text{if } x \notin J \end{cases} \tag{3}$$

If x falls within J, then $d = 0$ and the point is assigned a value of 1; for points outside J, d increases with distance from the nearest boundary, and the assigned value decreases accordingly toward 0. The final Aff_P and Aff_R are computed by averaging these per-point values. Aff_P

measures how close each predicted anomalous point is to the actual anomaly segment, and Aff_R measures how close the nearest prediction is to each point within the actual anomaly segment.

3.4 Threshold Methods

In anomaly detection, setting a threshold is required to classify the anomaly scores output by a model into normal and anomalous. We divided threshold methods into two categories: the Oracle approach, which references ground truth labels, and the Label-free approach, which determines thresholds based on the statistical properties of the data without labels.

Oracle Threshold. The Oracle approach selects the threshold that maximizes the F1-score by referencing the ground truth labels of the test data. Although it cannot be applied in operational environments where ground truth labels are unavailable, it serves as a baseline by providing the upper bound of performance that a model can achieve. We used two Oracle thresholds: Best F1, which maximizes point-wise F1, and Best $F1_{PA}$, which maximizes PA-F1.

Label-free Threshold. The Label-free approach sets thresholds based solely on the statistical properties of the data without ground truth labels, which can be applied to operational environments. We applied three Label-free thresholds. NDT (Nonparametric Dynamic Thresholding) [7] partitions the anomaly score sequence into windows and, within each window, selects the threshold that maximizes the relative decrease in mean and standard deviation when all values above it are removed. This enables the statistical identification of anomalous data points that deviate most from normal data. P_{95} and P_{99} use the 95th and 99th percentile values, respectively, from the anomaly score distribution of the training data as thresholds, determining anomalies based on extreme values of the normal score distribution under the assumption that the training data consists solely of normal data.

4 Experiments

4.1 Experimental Setup

In this study, we conducted two experiments. Figure 1 summarizes the overall experimental pipeline.

- **Exp1: Detection Performance by Model Type.** Exp1 is an experiment designed to measure the performance gap between traditional ML models and recent DL models that have achieved state-of-the-art performance on four ICS benchmark datasets. PCA and IForest as traditional ML models, and USAD and TranAD as DL models, were selected for comparison, with the dataset and preprocessing held constant and model type as the only independent variable. The hyperparameters of each model were kept at the default values from the original papers to compare out-of-the-box performance. Threshold-free metrics (AUROC, AUPRC) and distance-based metrics (Aff_P, Aff_R) were applied to compare performance from multiple perspectives.
- **Exp2: Impact of Threshold Selection.** This experiment examines how evaluation outcomes change when only the threshold method is varied without changing the underlying predictions. The anomaly scores from Exp1 were used directly, with the evaluation metric

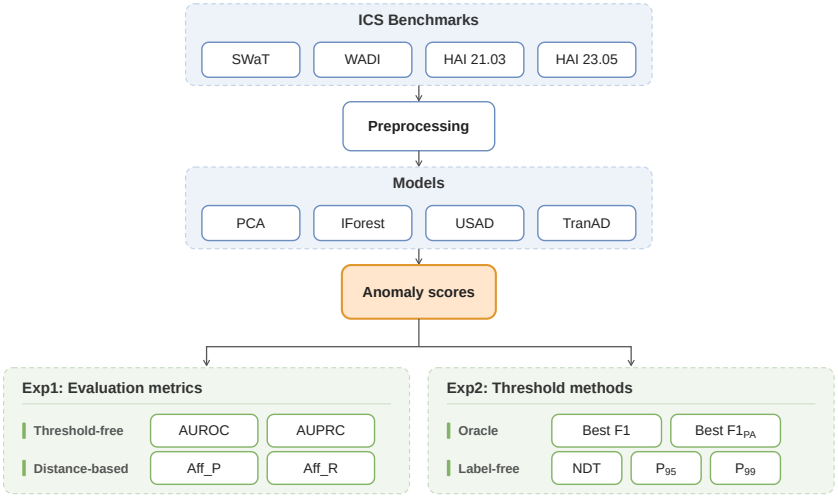


Figure 1: Overview of the experimental pipeline.

fixed at PA-F1 and the threshold method as the only independent variable. Five threshold methods were applied, including Oracle methods (Best F1, Best $F1_{PA}$) and Label-free methods (NDT, P_{95}, P_{99}), to compare the impact of threshold selection rather than the model itself.

The details of each dataset are summarized in Table 1.

Table 1: Details of the Benchmark Datasets

	SWaT	WADI	HAI 21.03	HAI 23.05
Features	51	123	79	86
# Train	495,000	784,571	921,603	896,400
# Test	449,919	172,803	402,005	284,400
Anomaly (%)	12.13	5.77	2.23	4.00

To ensure fairness, only basic preprocessing was applied to all datasets. Min-Max normalization was applied to all features; for WADI, 4 columns with entirely missing values were removed and the remaining missing values were imputed with the most recent available value (forward fill). No additional feature engineering was applied, and the hyperparameters of each model were kept at the default values of the publicly available code without dataset-specific tuning. For model training, all models except PCA involve randomness in their training and initialization, so a single run cannot guarantee stable results. Accordingly, all experiments were conducted with three different seeds (2026, 2027, and 2028), and all values reported in the text and tables are averaged over the three seeds. Figures 2 and 4 illustrate representative results from a single seed (2026).

Table 2: Model Performance Across 4 ICS Benchmarks

Dataset	Model	Threshold-free		Distance-based	
		AUROC	AUPRC	Aff_P	Aff_R
SWaT	PCA	0.743	0.244	**0.598**	0.544
	IForest	**0.838**	**0.732**	0.585	**0.782**
	USAD	0.789	0.682	0.542	0.415
	TranAD	0.341	0.088	0.533	0.456
WADI	PCA	0.506	0.053	0.499	0.590
	IForest	**0.734**	0.180	**0.542**	**0.899**
	USAD	0.503	**0.203**	0.499	0.590
	TranAD	0.521	**0.203**	0.499	0.590
HAI 21.03	PCA	**0.866**	**0.580**	**1.000**	0.098
	IForest	0.745	0.130	0.564	**0.939**
	USAD	0.742	0.350	0.997	0.218
	TranAD	0.627	0.204	**1.000**	0.060
HAI 23.05	PCA	**0.737**	**0.372**	**1.000**	0.033
	IForest	0.618	0.072	0.570	**0.863**
	USAD	0.599	0.153	**1.000**	0.033
	TranAD	0.580	0.115	**1.000**	0.014

4.2 Detection Performance by Model Type

In the first experiment, traditional ML models (PCA and IForest) achieved the highest performance in 15 out of 16 dataset-metric combinations. As shown in Table 2, the only case in which a DL model achieved the highest performance was on WADI-AUPRC, where both USAD and TranAD scored 0.203, differing from IForest (0.180) by only 0.023.

By dataset, IForest on SWaT and WADI, and PCA on HAI 21.03 and HAI 23.05, each achieved the highest performance in 6 out of 8 combinations. On HAI 23.05 in particular, most models other than PCA failed to achieve meaningful detection performance: under threshold-free metrics, PCA achieved the highest AUROC of 0.737, while the other three models remained at 0.580–0.618. These results confirm that the finding from prior work [17], which demonstrated on SWaT and WADI that PCA achieves higher performance than major DL models, holds equally on the HAI datasets.

In addition, even given identical anomaly scores, model performance varied considerably with the evaluation metric applied. On WADI, IForest achieved the highest AUROC of 0.734 but dropped to 0.180 on AUPRC, falling below the DL models. This is because, while the overall detection performance of DL models was lower, they achieved high precision in the high-confidence region. As AUPRC represents the area under the Precision-Recall curve, high precision in this region can elevate the overall AUPRC. Figure 2 supports this by presenting the Precision-Recall curves for each model on WADI, showing that DL models achieved high precision in the low-recall region. A similar pattern was found on HAI 23.05, where PCA achieved the highest AUROC of 0.737 but scored only 0.033 on Aff_R, and USAD and TranAD also scored 0.033 and 0.014, respectively. In contrast, IForest achieved the highest Aff_R of 0.863, demonstrating that model rankings can vary substantially depending on the choice of metric.

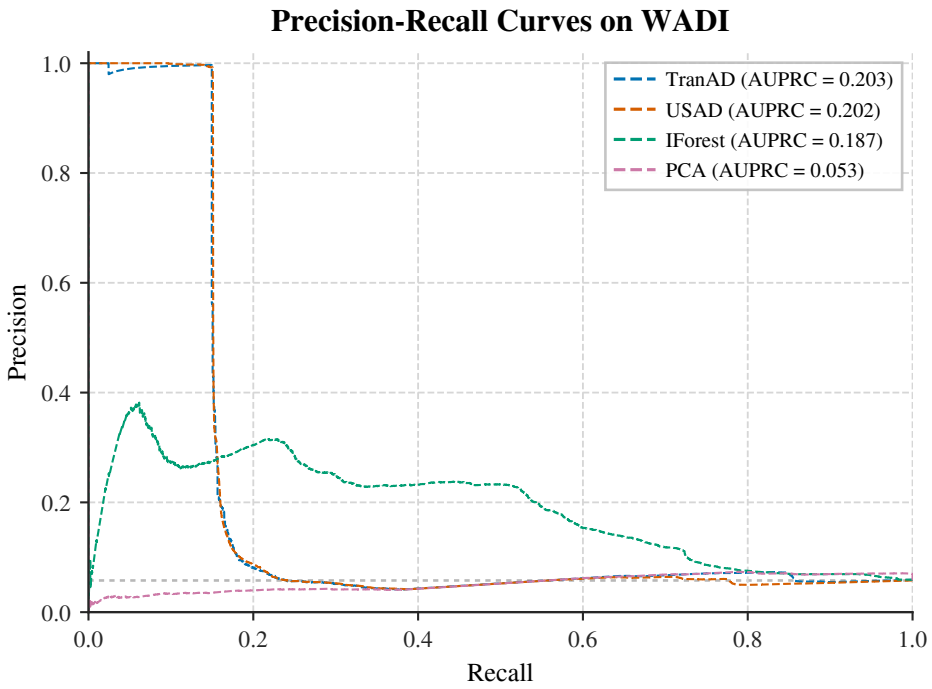


Figure 2: Precision-Recall curves of four anomaly detection models on WADI (single seed: 2026).

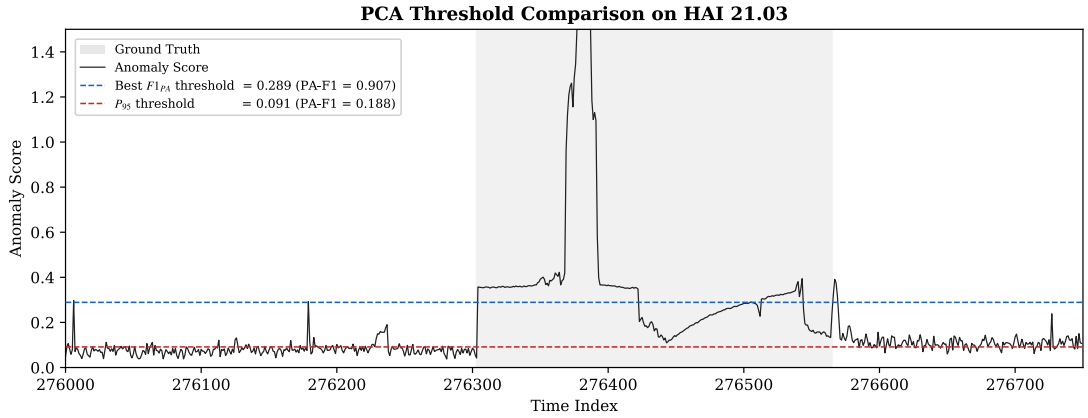


Figure 3: PCA anomaly scores on HAI 21.03 with two threshold lines. The same model yields PA-F1 of 0.907 or 0.188 depending solely on the threshold method.

4.3 Impact of Threshold Selection

In Exp2, when only the threshold method was changed for the same anomaly scores measured in Section 4.2, the PA-F1 score of a single model varied by up to 0.719. As shown in Table 3, PCA on HAI 21.03 dropped from 0.907 (Best $F1_{PA}$) to 0.188(P_{95}), a difference of 0.719, and USAD on SWaT dropped from NDT 0.823 to P_{95} 0.273, a decrease of 0.550. As Figure 3 illustrates

Table 3: PA-F1 Under Different Threshold Methods

Dataset	Model	Oracle		Label-free		
		Best F1	Best $F1_{PA}$	NDT	P_{95}	P_{99}
SWaT	PCA	0.564	0.565	0.753	0.304	0.325
	IForest	**0.803**	**0.846**	0.822	**0.507**	**0.839**
	USAD	0.801	0.805	0.823	0.273	0.282
	TranAD	0.250	0.351	**0.827**	0.284	0.289
WADI	PCA	0.110	0.119	0.491	0.165	0.133
	IForest	**0.473**	**0.672**	0.465	**0.480**	**0.608**
	USAD	0.279	0.299	0.513	0.143	0.133
	TranAD	0.279	0.279	**0.622**	0.143	0.136
HAI 21.03	PCA	**0.901**	**0.907**	0.201	0.188	**0.754**
	IForest	0.582	0.694	0.258	0.329	0.644
	USAD	0.774	0.787	**0.771**	0.255	0.448
	TranAD	0.599	0.613	0.612	**0.353**	0.502
HAI 23.05	PCA	**0.851**	**0.872**	0.408	**0.659**	**0.845**
	IForest	0.255	0.354	**0.415**	0.260	0.321
	USAD	0.316	0.333	0.390	0.193	0.261
	TranAD	0.227	0.325	0.399	0.245	0.282

for PCA on HAI 21.03, the two thresholds applied to the same anomaly score time series define markedly different decision boundaries. A lower threshold classifies more time points as anomalous, increasing recall but also increasing FP, while a higher threshold reduces FP at the cost of missing true anomalies. These results confirm that even when the model output remains identical, the choice of threshold alone can change the precision-recall balance and lead to substantial differences in the F1 score.

Furthermore, the best-performing model also changed with the threshold method. On HAI 21.03, PCA achieved the highest performance under the Oracle criterion, but USAD achieved the highest under NDT and TranAD under P_{95}. On SWaT as well, IForest achieved the highest under Best F1, whereas TranAD achieved the highest under NDT and IForest under P_{95}. This suggests that when reporting anomaly detection performance, a justification for the threshold method should be provided, and comparison across multiple methods is necessary.

Nevertheless, even with diverse threshold settings, the trend of higher ML model performance identified in Exp1 was generally maintained. In 16 out of 20 dataset-threshold combinations, traditional ML models achieved higher performance than DL models. DL models achieved the highest performance in four cases: three under NDT (SWaT, WADI, and HAI 21.03) and one under P_{95} on HAI 21.03. Given that DL models achieved generally lower performance than ML models under P_{95}/P_{99}, which are also Label-free methods, the reason why DL models performed favorably under NDT requires further analysis.

5 Discussion

Through two experiments, we showed that the performance of anomaly detection models measured on ICS benchmark datasets can be influenced as much by the evaluation metrics and

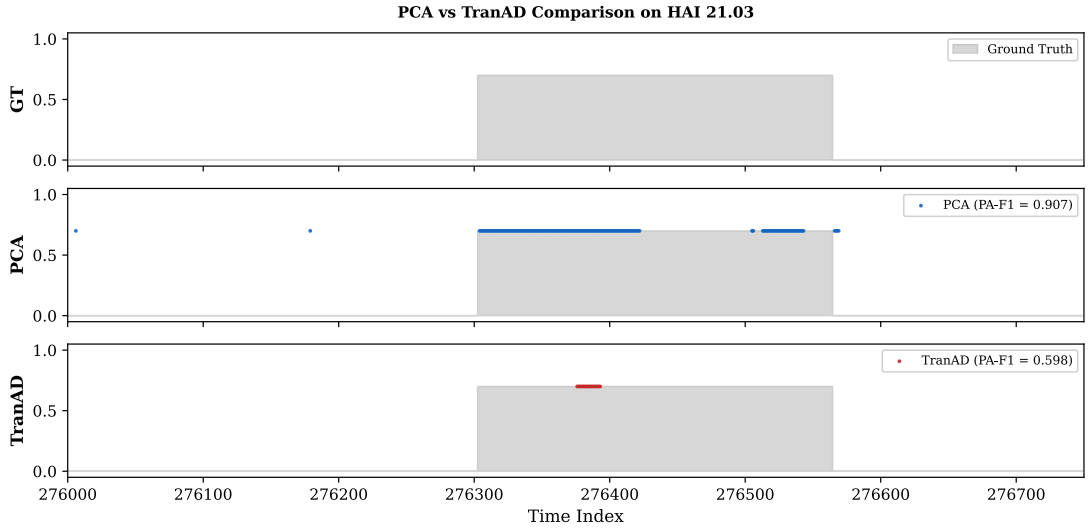


Figure 4: Detection results on HAI 21.03 under the Best $F1_{PA}$ threshold (single seed: 2026). PCA detects most ground truth segments, while TranAD misses substantial portions.

threshold methods used as by the model itself. When different evaluation metrics or threshold methods are applied to the same model output, the resulting performance gap between models can change, indicating that experimental results in existing ICS anomaly detection research are highly dependent on the evaluation methodology employed.

These results can be examined from two perspectives. First, from the model perspective, the finding from prior work [17] that traditional ML models outperform state-of-the-art DL models on SWaT and WADI was consistently confirmed on the HAI datasets also examined here. In the majority of dataset-metric combinations, ML models (PCA and IForest) achieved the highest performance, and on HAI 23.05 in particular, most models other than PCA failed to achieve meaningful detection performance. Second, from the evaluation perspective, even with the scores held constant, relative model performance changed with the evaluation metric or threshold method. On WADI, IForest achieved the highest AUROC but scored lower than DL models on AUPRC, and on HAI 23.05, PCA achieved the highest AUROC but IForest achieved the highest Aff_R, reversing the comparative result. In the threshold comparison experiment (Exp2) as well, the PA-F1 score of a single model differed by as much as 0.719 when only the threshold method was changed, and the model achieving the highest performance also varied by threshold method. Figure 4 illustrates this concretely on HAI 21.03: under the same threshold method (Best $F1_{PA}$), PCA detects most ground truth segments, whereas TranAD misses substantial portions.

These results indicate that performance reporting based on a single evaluation metric and a single threshold setting may not sufficiently reflect the true capability of a model. Given that comparative results themselves can change with the evaluation methodology, it is difficult to fairly compare models based on a single result alone. Therefore, when reporting performance in anomaly detection research, clear justification for the evaluation metric and threshold method should be provided, and where possible, comparison results across multiple metrics and threshold methods should be presented together. Furthermore, our results suggest that the appropriate evaluation metric and threshold method may differ depending on dataset characteristics, and

future research should systematically examine this and select suitable evaluation approaches.

5.1 Limitations

The following considerations should be taken into account when interpreting these results. First, in WADI-Affiliation, the Precision and Recall of three models (PCA, USAD, TranAD) were identical across the three models (0.499 and 0.590), respectively; under our experimental setup, this metric did not adequately distinguish the performance differences among these three models. Identifying the specific cause of this result requires further analysis. Second, on HAI 23.05, most models other than PCA failed to achieve meaningful detection performance, which appears to be related to the inherent detection difficulty of the dataset itself. Third, in Exp2, DL models achieved higher performance than ML models mainly under NDT among the Label-free thresholds, with one additional case under P_{95} on HAI 21.03, but we did not conduct an in-depth analysis of the underlying cause. However, the points mentioned above do not directly affect our core conclusions, and a more in-depth analysis is left for future work.

5.2 Future Work

Based on the impact of evaluation methodology identified here, future research may consider extensions in two directions. First, it is necessary to explore fair evaluation approaches that account for differences in anomaly score distributions among models. As illustrated by the WADI-Affiliation results discussed in Section 5.1, uniformly applying the same threshold to all models can cause differences in score distributions to directly influence evaluation outcomes, and follow-up research is needed to enable more reliable comparisons through score normalization or model-specific threshold selection. Additionally, we focused on evaluating anomaly detection as binary classification on ICS datasets; however, the SWaT, WADI, and HAI datasets all include labels for individual attack scenarios. Future work should examine whether improved evaluation methodologies can enhance not only binary anomaly detection but also the identification of and response to specific attack scenarios. Such research is expected to contribute to the timely detection and root-cause analysis of cyber attacks in ICS environments, as well as to improving the cyber-resilience of such systems [18]. The present evaluation methodology can also be extended to resilience-oriented aspects, such as detection delay, recovery behavior, and adaptation.

6 Conclusion

In ICS anomaly detection research, concerns about the reliability of reported performance have been persistently raised due to limitations in datasets and evaluation methodologies. We compared the performance of traditional ML models (PCA, IForest) and DL models (USAD, TranAD) on four ICS benchmark datasets (SWaT, WADI, HAI 21.03, HAI 23.05) using a variety of evaluation metrics (Exp1), and analyzed how performance changed across threshold methods for identical model outputs (Exp2). ML models matched or surpassed DL models in 15 of 16 dataset-metric combinations, and even with unchanged anomaly scores, relative model rankings shifted with the evaluation metric or threshold method. These findings indicate that reported anomaly detection performance is not determined solely by the model itself, and that the evaluation metrics and threshold methods used can directly influence the conclusions of model comparisons.

Our results indicate that ensuring the security and resilience of ICS environments requires anomaly detection research to focus not only on improving model performance but also on establishing robust evaluation methodologies. Beyond simple model performance comparison, a consistent and systematic examination of evaluation metrics and threshold methods must be integrated into the research process to ensure the validity of reported performance and the reliability of model comparisons. Future research can explore evaluation methodologies tailored to specific dataset characteristics. Additionally, it should be investigated whether improvements in evaluation reliability can contribute to threat detection and response in ICS security environments.

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